Strategic HR Analytics & Employee Attrition Predictor

Executive Overview, KPI Metric Explanations, Feature Guide, and Word-for-Word Presenter Script

1. Executive Overview: What is PulseHR & Why Does It Matter?

Employee turnover (when employees resign and leave an organization) is among the most costly problems modern companies face. Replacing a single skilled employee typically costs **50% to 200% of their annual salary** in recruitment fees, onboarding delays, and lost team momentum. **PulseHR** transforms Human Resources from *reactive* (learning someone is unhappy only during an exit interview) into *proactive* (identifying flight risk months ahead and providing managers with targeted retention strategies).

Key Takeaway for Presenters: PulseHR answers three fundamental business questions:
1. **What is happening right now?** (Live Headcount, Attrition Rate %, Average Compensation, Job Satisfaction).
2. **Why are people leaving?** (Overtime burnout, compensation disparities, early-tenure turnover).
3. **Who is at risk next, and how do we retain them?** (AI predictive flight risk scoring and tailored HR action plans).

2. Essential Terminology Glossary

Use these standardized industry terms during your presentation to speak with confidence and clarity:

Term	Plain-English Meaning	How It Applies in the Platform
Attrition Rate (%)	The percentage of the workforce that resigns over a period.	Shown as a primary KPI card (14.2%). Healthy industry benchmarks stay under 15%.
Flight Risk	The calculated likelihood (0%–100%) that an employee will leave.	Visualized on the Radial Gauge Meter in Tab 2 and in the 360° Profile Inspector.
Tenure	How many years an employee has worked at the company.	Reveals vulnerable turnover zones, particularly during the first 0–2 years.
Hygiene vs. Motivator Factors	Herzberg's HRM theory: Hygiene factors (salary, overtime) cause turnover if bad; Motivators (satisfaction, growth) drive retention.	Tested across both dimensions: compensation/overtime (hygiene) and satisfaction score (motivator).
Random Forest Classifier	An ensemble AI algorithm combining decision trees to predict flight risk accurately without bias.	Powers the interactive prediction form in Tab 2 with balanced class weighting.
Feature Importance	A ranking showing which factors have the highest impact on turnover decisions.	Visualized as a horizontal bar chart, proving OverTime is the #1 churn driver.

3. Deep Dive: Top Executive KPI Cards Explained

At the top of the dashboard, five high-visibility metric cards summarize the current organizational health. Here is what each number represents, why it matters to leadership, and exactly what the presenter should say:

KPI Card	Current Value & Meaning	Strategic Business Importance	What the Presenter Should Say
■ **Total Headcount**	**1,200** *(100.0% of org)* The total number of employee records currently matching the active filter criteria.	Represents organizational size and statistical sample. Dynamic: if you filter by 'Sales', it automatically reflects the Sales headcount.	*"This card gives us our active headcount baseline—1,200 employees across all divisions. As we filter by department or income, it adjusts in real time."*
■ **Attrition Rate**	**14.2%** *(171 Exited Staff)* The percentage of the total workforce that has resigned or separated.	**The #1 HR vital.** Industry standard for healthy turnover is under 15%. PulseHR color-codes this card: Green (<12%), Amber (12–20%), Red (>20%).	*"Our current attrition rate is 14.2%, accounting for 171 departures. It is color-coded amber to signal that turnover is near the caution threshold."*
■ **Avg. Monthly Income**	**$10,982** *(Per Employee / Mo)* The mean gross monthly compensation across all roles and experience tiers.	Establishes the internal compensation benchmark. When compared to the lower pay of departed staff in Tab 1, it proves salary disparity drives resignations.	*"Our average monthly compensation stands at $10,982. This acts as our salary baseline to verify whether lower pay is driving talent poaching."*
■ **Avg. Job Satisfaction**	**2.54 / 4.0** *(Status: Moderate)* Average employee morale score from pulse surveys, rated 1 (Low) to 4 (Very High).	Measures sentiment and engagement. In HR theory, satisfaction is the key motivator factor. When satisfaction drops below 2.0, flight risk skyrockets.	*"Job satisfaction averages 2.54 out of 4, rated as Moderate. In retention theory, satisfaction is the core motivator preventing voluntary exits."*
■■ **OverTime Load**	**29.8%** *(Active Overtime Ratio)* The proportion of the workforce regularly clocking mandatory or extra hours.	**The primary burnout thermometer.** When nearly 30% of employees work overtime, chronic fatigue sets in. In our ML model, this is the #1 churn driver.	*"Nearly 30% of our staff is working overtime. This is a critical indicator of workload fatigue and the primary predictor of upcoming turnover."*

Real-Time Dynamic Sidebar Filter Controls

On the left sidebar, three controls allow users to slice and dice the workforce: **1. Department** (multi-select for R&D;, Sales, HR), **2. OverTime Commitment** (segmented toggle: All Staff, Overtime Only, Standard Hours), and **3. Monthly Income Slider** (dynamic range). Clicking **'■ Reset All Filters'** instantly restores the full organization view.

4. Step-by-Step Feature Walkthrough: The Three Functional Tabs

PulseHR divides workforce analysis into three focused operational views:

Tab View	Visual Elements & Charts	What It Tells the Presenter & Audience
■ **Tab 1: Workforce Insights & Analytics**	• **Departmental Attrition:** Grouped bar chart comparing active vs exited staff. • **Monthly Income Box Plot:** Shows salary distribution and median income gap between stayers and leavers. • **Job Satisfaction Bar:** Turnover percentage across satisfaction tiers 1 to 4. • **Tenure vs. Overtime Scatter:** Correlates company years, salary, and overtime. • **HRM Takeaways Callout:** Summarizes macro-level retention insights.	Provides the **macro-level 'Why'**. Proves statistically that: (1) departed employees had significantly lower median incomes, (2) turnover peaks in years 0–2 (early tenure churn), and (3) employees on overtime experience twice the turnover rate of standard-hour workers.
■ **Tab 2: Attrition Risk Predictor (ML)**	• **Employee Profile Simulation Form:** Sliders for Age, Tenure, Overtime toggle, Monthly Income, and Satisfaction. • **Radial Gauge Meter:** Displays exact probability score (0% to 100%). • **Risk Badges:** Low (<30%, Green), Moderate (30–60%, Amber), High (>60%, Red). • **Feature Importance Chart:** Ranks top churn drivers. • **Actionable Retention Interventions:** Contextual management playbooks.	Provides the **predictive 'Who & What to do'**. Simulates any employee's flight risk on the fly. Rather than just giving a score, it provides specific retention actions (e.g. rebalance overtime, schedule skip-level 1-on-1, benchmark compensation, assign mentor).
■ **Tab 3: Employee Records & Search**	• **Universal Search Bar:** Instant query across Employee ID, Job Role, and Department. • **Talent Facets:** Filter by Attrition status, Job Role, Satisfaction, Tenure. • **Multi-Column Sorting:** Sort by Income, Tenure, Morale, Age, ID. • **Live Cohort KPIs:** Summary metrics for the filtered group. • **Individual 360° Profile Inspector:** Pulls up full file and calculates real-time ML risk. • **CSV Export:** One-click audit report download.	Provides the **operational execution tool** for HR managers. Allows looking up specific individuals or cohorts (e.g. all R&D; engineers with low satisfaction) and downloading clean CSV reports for leadership meetings.

5. Word-for-Word Presenter Script (5-Minute Demonstration Flow)

Use this exact script during live presentations or video demos. Pair each spoken line with the screen action:

Step 1: The Hook & Introduction [0:00 - 0:45]

Action on Screen: *Stand on the main dashboard showing the Hero Banner and Top KPI cards.*

*"Hello everyone. Today, I'm presenting **PulseHR**, a modern Talent Analytics and Attrition Predictor platform.*
*In business, employee turnover is an expensive crisis—replacing a talented staff member costs between 50% and 200% of their salary in recruitment fees, training delays, and lost productivity. Most companies only learn an employee is unhappy during their exit interview—when it is already too late. PulseHR changes this: it moves HR from **reactive firefighting** to **proactive retention** using analytics and machine learning."*

Step 2: Executive Health KPIs & Live Dynamic Filters [0:45 - 1:45]

Action on Screen: *Point out the 5 KPI cards. Then adjust the sidebar Department filter and Monthly Income slider.*

"At the top of our dashboard, we see five key organizational vitals: a Total Headcount of 1,200 employees, an overall Attrition Rate of 14.2% (representing 171 departures), an average monthly salary of $10,982, average job satisfaction of 2.54 out of 4, and an OverTime load of 29.8%.
*On the left sidebar, we have dynamic filter controls. Watch what happens when I filter down to the **Sales** department and adjust our salary slider. Every single metric card, chart, and data record on the dashboard recalculates instantly. This lets leadership evaluate specific departments or salary bands in seconds."*

Step 3: Deep Workforce Insights (Tab 1) [1:45 - 2:45]

Action on Screen: *Click Tab 1: 'Workforce Insights & Analytics'. Point to the box plot and the overtime scatter plot.*

*"In our **Workforce Insights** tab, we explore **why** people leave.*
Looking at our Monthly Income Box Plot, we observe a noticeable compensation gap: employees who left earned significantly lower median wages than those who stayed, confirming that salary competitiveness is critical.
Next, looking at our Tenure versus Overtime scatter plot, we discover two vital patterns: turnover peaks during the first 0 to 2 years—known as early-career churn—and employees on overtime experience nearly double the turnover rate of standard-hour workers, proving that chronic burnout drives resignations."

Step 4: Live AI Attrition Risk Predictor Demo (Tab 2) [2:45 - 4:00]

Action on Screen: *Click Tab 2. In the form, set Age=32, Tenure=2, OverTime=Yes, Monthly Income=$4,200, Satisfaction=Low (1). Click 'Predict Attrition Risk'.*

*"Now let's examine our most powerful feature: the **Attrition Risk Predictor** powered by a machine-learning Random Forest model.*
Imagine an HR manager wants to evaluate a 32-year-old engineer who has been with us for 2 years, earns $4,200 a month, works overtime, and reported low satisfaction.
*I'll click **'Predict Attrition Risk'**... and instantly, our AI calculates an **81% Flight Risk (High Risk Tier)** on the radial gauge.*
*Crucially, PulseHR doesn't just deliver a number; it provides **Actionable Retention Interventions**. The system recommends: workload rebalancing to mitigate overtime burnout, a structured 1-on-1 skip-level meeting to address sentiment, and a salary parity review. This gives managers a precise playbook to retain that talent before they submit their notice."*

Step 5: Talent Search, 360° Profile & CSV Export (Tab 3) [4:00 - 5:00]

Action on Screen: *Click Tab 3. Type 'Research' in the search bar. Open 'Inspect Individual Employee 360° Profile'. Point to 'Export Searched Cohort to CSV'.*

*"Finally, in Tab 3, we have our **Employee Directory & Advanced Talent Search**.*
HR managers can use our universal search bar or filter facets to isolate cohorts—such as employees with low satisfaction.
Above the table, live Cohort KPI cards calculate the exact average salary and turnover rate of the searched group.
*We can also open the **Individual Employee 360° Profile Inspector** to review any team member's full compensation and real-time flight risk score.*
*And with one click on **'Export Searched Cohort to CSV'**, HR can download a clean report for executive meetings."*

Closing & Q&A; Readiness [5:00 - 5:30]

"In summary, PulseHR bridges data science and human capital management to protect an organization's most valuable asset—its people. Thank you, and I look forward to your questions!"

6. Anticipated Questions & Winning Answers for the Presenter

Be prepared for these common questions from professors, interviewers, or managers:

Question	Recommended Winning Answer
"Where does the employee dataset come from?"	*"The data is statistically generated using NumPy with a fixed random seed. It mirrors the realistic distributions of standard industry benchmarks (like the IBM HR Analytics dataset), incorporating realistic correlations like overtime stress and early-tenure turnover."*
"How does the Machine Learning model calculate flight risk?"	*"We train a Random Forest Classifier with balanced class weights using Scikit-Learn. It evaluates key features—Overtime, Monthly Income, Job Satisfaction, Age, and Company Tenure—and generates a calibrated probability score (0% to 100%) indicating how closely the employee matches historical departure patterns."*
"Why is OverTime the strongest predictor of attrition?"	*"In HRM organizational psychology, sustained overtime leads directly to work-life conflict and burnout. In our feature importance ranking, Overtime emerged as the single highest-weighted flight risk driver, proving that compensation alone cannot offset chronic workload fatigue."*
"Why does attrition peak in the first 0 to 2 years of tenure?"	*"This reflects early-career onboarding vulnerability. New hires who experience poor onboarding, unclear expectations, or role misalignment will quickly look for opportunities elsewhere. After year 3, employees typically form stronger organizational ties."*
"How would a real-world enterprise integrate PulseHR?"	*"In a live corporate environment, PulseHR connects directly to HR Information Systems (HRIS) like Workday, BambooHR, or SAP SuccessFactors via API. It runs quarterly retention scans and alerts department heads when high-performing staff enter high-risk flight zones."*